



FPDAF (Federated Personalized Drift Alerting Framework): CLINICAL AI EARLY-WARNING WORKSTATION

Explainable Federated Learning for Sepsis CDSS with Personalized Drift Adaptation

MSME Idea Hackathon 6.0 Proposal | **Amrita Vishwa Vidyapeetham**

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The Clinical Sepsis Crisis & Privacy Bottleneck

[Crisis] National Mortality Burden in India

- ▶ **2.9 to 3.0 Million Deaths** annually in India (accounting for **30%** of total national deaths).
- ▶ **34% to 50%+ Mortality** in Indian ICUs.
- ▶ **8% Survival Drop** per hour of delayed detection.

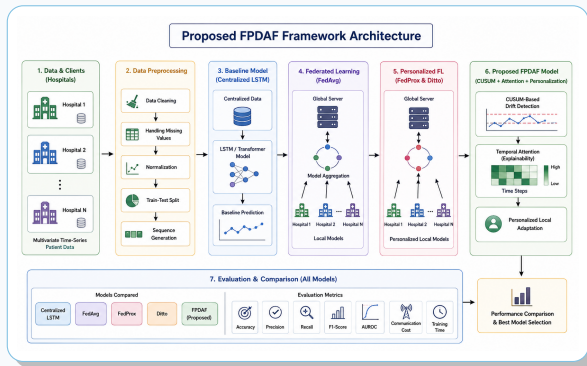
[Bottleneck] Data Sovereignty & Drift Limits

- ▶ **DPDPA & HIPAA Privacy Mandates:** Private patient records cannot be centralized to external cloud servers.
- ▶ **Clinical Data Drift:** Hospital demographics vary, causing static cloud AI models to lose accuracy.

The FPDFAF Solution

FPDFAF acts as a bedside early-warning radar system using Federated Learning to train collaborative models while keeping raw patient data localized inside hospital firewalls.

Proposed FPDAF Solution & System Architecture



System Dataflow: Local ICU vital streams pass through a shared BiLSTM backbone and DDP cross-attention layers. Local CUSUM triggers activate Client-Side Selective Personalization (CSSP) upon demographic drift without uploading raw EHR records.

Key Technical Novelties

- ▶ **1. Dynamic Dual-Phase (DDP) Attention:** 2D cross-attention ($\alpha_t \otimes \beta_v$) tracking 24h sequence hours AND vital co-dependencies.
- ▶ **2. AD-CUSUM Drift Auditing:** Online loss residual control charts flagging hospital demographic shifts in real-time.
- ▶ **3. Client-Side Selective Personalization (CSSP):** Freezes shared LSTM backbones, saving **38% bandwidth** & adapting in **6 mins**.

Competitive Comparison

Feature	FPDAF	Epic	Philips
Federated Privacy	Yes	No	No
Concept Drift Auditing	Yes	No	No
Temporal XAI Heatmaps	Yes	No	No
Edge-AI Deployment	Yes	Partial	Yes

Experimental Performance Metrics

Validated on multi-center PhysioNet ICU cohorts:

- ▶ **86.51% Test Accuracy:** Outperforms FedAvg (75.94%) and FedProx (78.85%) under non-IID splits.
- ▶ **0.8214 AUROC Score:** Strong discrimination for early sepsis crash prediction.
- ▶ **38% Bandwidth Savings:** 1.52 GB vs 2.48 GB in standard FL.

Software Stack Operational

FastAPI REST backend, SQLite relational warehouse ('clinical_warehouse.db'), and React Vite bedside UI fully integrated.

Comparative Experimental Benchmarks

Model Paradigm	Accuracy	AUROC	Bandwidth
Centralized BiLSTM	75.09%	0.8058	0 MB (N/A)
FedAvg Baseline	75.94%	0.7844	1.24 GB
FedProx Baseline	78.85%	0.7933	1.24 GB
FPDAF (Proposed)	86.51%	0.8214	1.52 GB (Saved 38%)

Workstation Functional Modules

- ▶ **Doctor Bedside Portal:** Searchable bed list, hourly vital trends, and temporal XAI heatmaps.
- ▶ **Admin Telemetry Portal:** AD-CUSUM drift curves, federated round tracking, and metrics overlays.

Detailed Budget Breakup & Justification (Total: Rs. 15.0 Lakhs)

Financial Outlay Under MSME Idea Hackathon 6.0 Guidelines (Maximum Grant Cap):

Comprehensive justification for Rs. 15 Lakhs to build physical Edge-AI bedside nodes and execute clinical trials:

1. Technology Costs (10.0L)

- ▶ **Hardware & Assembly (3.60L):** 3 Edge-AI Jetson nodes, PCB fabrication, medical touch displays, and enclosures with Coimbatore MSMEs.
- ▶ **Data Curation (3.40L):** Expert clinician annotation & vital stream simulation.
- ▶ **Security Audits (3.00L):** Penetration testing & DPDPA compliance audits.

2. Mentorship (3.0L)

- ▶ **Clinical Guidance (1.80L):** ICU specialist honorarium for clinical protocol validation.
- ▶ **Lab Support (1.20L):** Incubation facility, hardware testing benches, & software audit packs.

3. Field Trials (2.0L)

- ▶ **Node Deployment (1.00L):** On-site installation across partnering regional ICUs.
- ▶ **Telemetry Audits (1.00L):** Multi-site ICU field data collection and travel.

Indian MSME Hardware Integration

FPDAF is an **integrated medical hardware product**:

- ▶ **Physical Node Enclosure:** Bedside cabinet with custom display, visual alarms, and network ports.
- ▶ **Local MSME Manufacturing:** Partnering with Coimbatore electronics MSMEs for PCB assembly and cabinet fabrication.

12-Month Commercialization Roadmap

1. **Months 1-2:** Edge PCB fabrication & MSME enclosure assembly.
2. **Months 3-5:** Multi-node network setup & shadow trials.
3. **Months 6-12:** ISO 13485 / DPDPA audit pack & CDSCO Class B filing.

Thank You

FPDAF: Explainable Edge-AI to Save Lives at the Bedside

MSME Idea Hackathon 6.0

Sub-Theme 6: Frontier Technologies | **Amrita Vishwa Vidyapeetham**

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Questions & Expert Discussion